```c
1WA24CS226
#include <stdio.h>
#define MAX 10
#define INF 9999

int gcd(int a,int b){ return b?gcd(b,a%b):a; }
int lcm(int a,int b){ return a*b/gcd(a,b); }

int main(){
    int n,i,t,hp;
    int b[MAX],p[MAX],d[MAX],rem[MAX],ad[MAX];

    printf("Enter number of processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++){
        printf("Enter burst, deadline, period for T%d: ",i+1);
        scanf("%d%d%d",&b[i],&d[i],&p[i]);
        rem[i]=0;
    }

    hp=p[0];
    for(i=1;i<n;i++) hp=lcm(hp,p[i]);

    printf("\nPID\tBurst\tDeadline\tPeriod\n");
    for(i=0;i<n;i++)
        printf("%d\t%d\t%d\t\t%d\n",i+1,b[i],d[i],p[i]);

    printf("\nScheduling occurs for %d ms\n\n",hp);

    for(t=0;t<hp;t++){
        for(i=0;i<n;i++)
            if(t%p[i]==0){
                rem[i]=b[i];
                ad[i]=t+d[i];
            }

        int sel=-1,min=INF;
        for(i=0;i<n;i++)
            if(rem[i]>0 && ad[i]<min){
                min=ad[i];
                sel=i;
```

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        for(i=0;i<n;i++)
            if(rem[i]>0 && ad[i]<min){
                min=ad[i];
                sel=i;
            }

        if(sel==-1) printf("%dms : CPU is idle.\n",t);
        else{
            printf("%dms : Task %d is running.\n",t,sel+1);
            rem[sel]--;
        }
    }
}
```

```
Enter number of processes: 3
Enter burst, deadline, period for T1: 2 4 5
Enter burst, deadline, period for T2: 3 7 20
Enter burst, deadline, period for T3: 2 8 10

PID     Burst   Deadline        Period
1       2       4               5
2       3       7               20
3       2       8               10

Scheduling occurs for 20 ms

0ms : Task 1 is running.
1ms : Task 1 is running.
2ms : Task 2 is running.
3ms : Task 2 is running.
4ms : Task 2 is running.
5ms : Task 3 is running.
6ms : Task 3 is running.
7ms : Task 1 is running.
8ms : Task 1 is running.
9ms : CPU is idle.
10ms : Task 1 is running.
11ms : Task 1 is running.
12ms : Task 3 is running.
13ms : Task 3 is running.
14ms : CPU is idle.
15ms : Task 1 is running.
16ms : Task 1 is running.
17ms : CPU is idle.
18ms : CPU is idle.
19ms : CPU is idle.

Process returned 0 (0x0)   execution time : 30.455 s
Press any key to continue.
```